

Magnetic Nanoparticle Quantification Using a Delta-E Magnetoelectric Sensor

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Abstract—In biomedical applications, magnetic nanoparticles (MNPs) play a key role in target drug delivery, hyperthermia, and contrast imaging. Here, delta-E magnetoelectric field sensors offer a promising platform for real-time localization and quantification of MNPs in an unshielded environment. To investigate this, the AC susceptibility of different concentrations of 50 μL MNP suspensions was measured at an excitation frequency of 180 Hz. The homogeneous 100 μT excitation magnetic field was generated by two rectangular coils in a Helmholtz configuration with a side length of 47 cm. The phase shift between the excitation field and the sensor response was recorded as a function of iron concentration. An increasing trend of phase shift with concentration was observed. Under the assumption of a Debye-type relaxation model, this trend is qualitatively related to an increase in the complex magnetic susceptibility. The results indicate that delta-E magnetoelectric sensing is sensitive to changes in MNP rotational dynamics in unshielded environments, supporting its potential for future biomedical sensing applications.

Index Terms—AC susceptibility, biosensing, Brownian relaxation, delta-E effect, magnetic nanoparticles

I. INTRODUCTION

Magnetic nanoparticles (MNP) are widely used in biomedical applications like contrast imaging and targeted drug delivery [1]. To localize and characterize MNP, they are excited via pulsed or continuous AC magnetic fields. Therefore, a variety of magnetic resonance imaging [2], magnetic particle imaging [3], magnetic particle spectroscopy [4], magnetorelaxometry [5] and AC susceptibility (ACS) [6] systems have been developed.

Conventional ACS systems are typically based on excitation and detection coils [7], [8]. More advanced systems employ fluxgate-based [9], [10], optically pumped magnetometer-based [11] and superconducting quantum interference device-based [12], [13] ACS measurements. In addition, magnetoelastic and magnetoelectric sensors have emerged as promising alternatives for MNP detection [14], [15]. Magnetoelastic sensors exploit the coupling between magnetic and elastic properties. In particular, delta-E effect sensors utilize magnetic-

field-dependent variations of the elastic modulus, resulting in resonance frequency shifts. [16], [17].

Our recent work demonstrated the quantification of MNPs using a surface acoustic wave (SAW)-based magnetoelastic sensor [18]. In the present work, a delta-E effect magnetoelectric sensor is employed in combination with a Helmholtz coil configuration to provide a homogeneous excitation field and to investigate phase-resolved MNP detection under unshielded conditions.

II. METHODS

A. Rotational Dynamics of Magnetic Nanoparticles

The rotational dynamics of MNP are governed by Brownian relaxation, corresponding to the rotation of the entire particle in a fluid, and Néel relaxation of the internal magnetic moment [19], [20]. For thermally blocked particles, Brownian relaxation dominates, with a characteristic time

$$\tau_B = \frac{3\eta V_H}{k_B T}, \quad (1)$$

where η is the dynamic viscosity, V_H the hydrodynamic volume, k_B the Boltzmann constant, and T the absolute temperature. In ACS measurements, the response of an MNP ensemble to an oscillating magnetic field is described by the complex susceptibility $\chi(\omega)$, with $\omega = 2\pi f$, the in-phase magnetic response $\chi'(\omega)$, and the out-of-phase component $\chi''(\omega)$ associated with dissipative losses. Within the Debye relaxation model for Brownian dynamics, the frequency-dependent susceptibility is given by

$$\chi' = \frac{\chi_0}{1 + (\omega\tau_B)^2}, \quad \chi'' = \frac{\chi_0 \omega\tau_B}{1 + (\omega\tau_B)^2}, \quad (2)$$

with the static susceptibility

$$\chi_0 = \frac{\mu_0 N \mu^2}{3k_B T}, \quad (3)$$

where μ_0 is the vacuum permeability, N the particle number density, and μ the magnetic moment of a single particle [21]. The maximum of $\chi''(\omega)$ occurs at $\omega = 1/\tau_B$, corresponding to the characteristic relaxation frequency f_c . Consequently,

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ACS measurements provide access to the effective relaxation behavior, which is influenced by hydrodynamic size, medium viscosity, and particle concentration [8], [22].

B. AC Susceptibility Measurement Setup

As shown in Fig. 1, the excitation system consists of two identical coils arranged vertically in a near-Helmholtz configuration. Each coil has a rectangular shape with rounded corners and comprises 18 turns of 1.5 mm diameter enamelled copper wire (IEC 60317-13, Grade 2), following the design proposed in [23]. The coils are separated by a center-to-center distance of 235 mm. The magnetic field homogeneity was experimentally characterized using an FM 302 teslameter equipped with an AS-LTM probe. At an excitation field amplitude of 1 mT, the field deviation within a $100 \times 50 \times 80 \text{ mm}^3$ volume was below 1%. To minimize magnetic interference, all screws and threaded components are made of aluminum or titanium. An acrylic glass plate between the coils serves as a support for the 3D-printed polylactic acid sample holder containing the MNP sample. The printed circuit board (PCB) carries a wire-bonded microelectromechanical system (MEMS) chip containing multiple ΔE -effect sensor elements and is positioned directly beneath the sample. In this work, sensor 9.1, as designated in [24], was employed. The sensor was biased by a permanent magnet generating a magnetic field of approximately 450 μT . The distance between the sensor and the MNP sample was on the order of a few millimeters.

Fig. 2 illustrates the measurement setup. An 180 Hz excitation signal generated in MATLAB was amplified and applied to Helmholtz coils to generate a homogeneous magnetic field of 100 μT . The delta-E effect sensor was operated at its resonance frequency of 343.33 kHz using a Zurich Instruments MFLI lock-in amplifier. The demodulated sensor response and the reference signal were synchronously acquired and processed in MATLAB to extract the MNP signal.

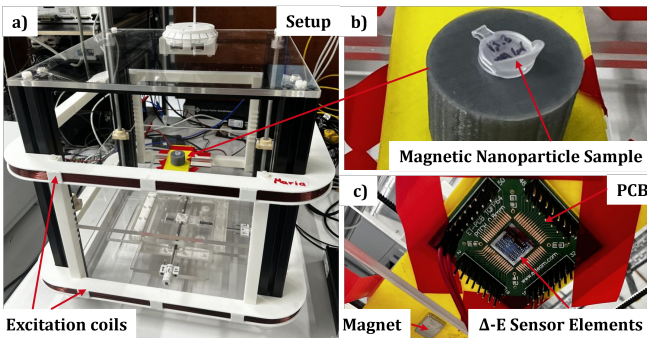


Fig. 1: a) Experimental setup, including the excitation coils, the sample holder, and the delta(Δ)-E sensor. b) MNP sample holder placed on top of the sensor. c) Permanent magnet for DC-bias field generation and PCB including the MEMS chip with the delta-E sensor elements.

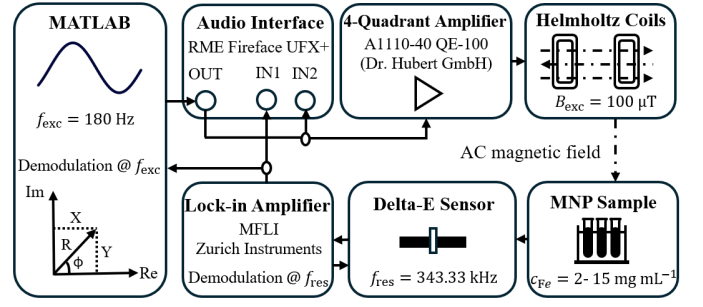


Fig. 2: Experimental setup and signal flow used for the detection of the resonance frequency shift induced in the delta-E sensor by the excited magnetic nanoparticles.

C. Samples and Measurement Procedure

At first, the dynamic susceptibility of 50 μL plain bionized nanorerrite MNP (micromod Partikeltechnologie GmbH, 100 nm) with a hydroxyethyl starch surface and a stock iron concentration $c_{\text{Fe}} = 15 \text{ mg mL}^{-1}$ was investigated using a commercial DynoMag® ACS system (RI.SE research Institutes of Sweden AB, version 1.13). The AC excitation field was set to a field strength of $B_{\text{AC}} = 500 \mu\text{T}$ with a frequency range from 10 Hz up to 10 kHz and the MNP susceptibility measured at 30 frequency points.

Secondly, the MNP were diluted in DI water to $2\text{--}15 \text{ mg mL}^{-1}$. The samples were placed and removed by hand into the sample holder on top of the delta-E sensor, and time-dependent measurements at $f_{\text{exc}} = 180 \text{ Hz}$ were conducted. Each measurement was repeated three times. The measurement protocol was structured into four phases:

- *Baseline phase*: empty sample holder ($t < 17.5 \text{ s}$, $t > 42.5 \text{ s}$), serving as noise reference.
- *Loading phase*: sample placement and stabilization ($17.5 \text{ s} \leq t < 22.5 \text{ s}$).
- *Signal phase*: steady-state response with sample present ($22.5 \text{ s} \leq t \leq 37.5 \text{ s}$).
- *Removal phase*: sample removal and post-baseline measurement ($37.5 \text{ s} \leq t < 42.5 \text{ s}$).

All signals were acquired in terms of in-phase X and quadrature Y components, from which amplitude R and phase ϕ were computed as described in Fig. 2. The system response was evaluated by comparing the steady-state signal during the signal phase to the combined pre- and post-measurement baseline, enabling extraction of the signal shift induced by the presence of magnetic nanoparticles.

III. RESULTS AND DISCUSSION

Fig. 3 shows $\chi''(\omega)$ of the MNP measured with the DynoMag® as a function of excitation frequency. From the peak position, a characteristic relaxation frequency of about $f_c \approx 180 \text{ Hz}$ is obtained. Based on this, $f_{\text{exc}} = 180 \text{ Hz}$ was chosen for the delta-E sensor measurements.

At that frequency, the delta-E signal phase exhibits a clear shift upon the introduction of the MNP sample ($c_{\text{Fe}} = 15 \text{ mg mL}^{-1}$). After sample removal, it returns to its

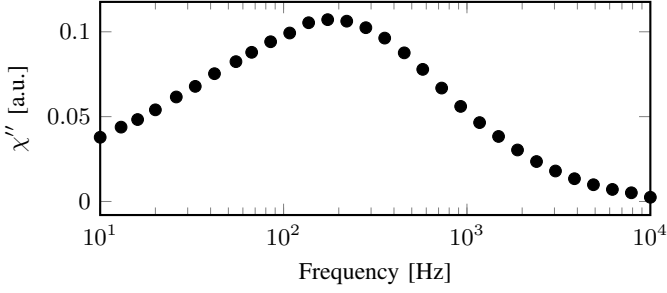


Fig. 3: Frequency-dependent imaginary susceptibility χ'' for a 10 mg/mL sample measured with the DynoMag®.

baseline value of -24 deg, which corresponds to the intrinsic phase offset of the 100 μ T excitation field without magnetic loading, as shown in Fig. 4. This behavior is consistent with a phase-sensitive response of the sensor system to the presence of magnetic nanoparticles and was observed in all three repetitions of the measurement.

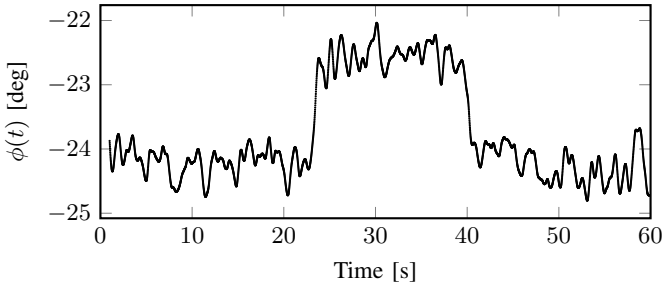


Fig. 4: Time-dependent phase response $\phi(t)$ for the first repetition of the 15 mg/mL sample.

To quantify the response, the phase difference was defined as $\Delta\phi = \phi_{\text{MNP}} - \phi_{\text{noise}}$, where ϕ_{noise} corresponds to the baseline phase. As shown in Fig. 5, $\Delta\phi$ increases approximately linearly with concentration across all three measurements. A zero-intercept linear fit yields a strong agreement with the data ($R = 0.984$), consistent with the Debye relaxation model described in II-A. Remaining deviations from linearity are attributed to noise and pipetting uncertainties.

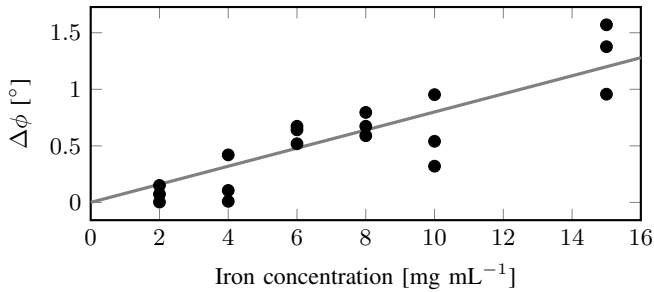


Fig. 5: Phase response $\Delta\phi$ as a function of iron concentration at 180 Hz. Each point corresponds to an individual repetition. A zero-intercept linear fit ($R = 0.984$) is shown as a gray line.

Although a linear trend is observed, the overall sensitivity for the delta-E sensor remains limited, as shown by the average signal to noise ratio (SNR) ratio, which increases with c_{Fe} as depicted in Fig. 6. It should be noted that only the 15 mg/mL sample yields a response exceeding the 3σ noise threshold, indicating that reliable detection is currently limited to relatively high concentrations. The SNR is evaluated solely based on the phase response $\Delta\phi$, combining contributions from both in-phase and quadrature components, making it more sensitive to variations in the underlying signal.

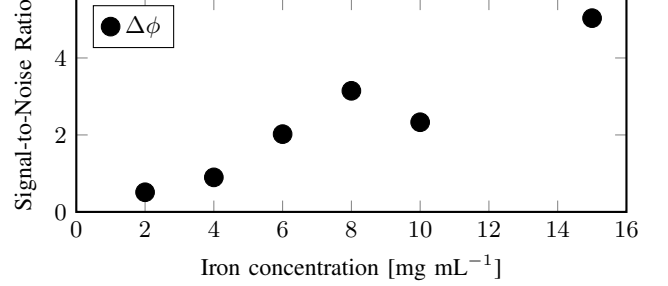


Fig. 6: Signal-to-noise ratio based on the phase response $\Delta\phi$ as a function of iron nanoparticle concentration.

In general, the delta-E sensor has a restricted operational bandwidth, which exhibits optimal performance up to approximately $f_{\text{exc}} = 400$ Hz. Consequently, the present system is constrained to low-frequency ACS measurements, which limits both sensitivity and flexibility in excitation schemes, compared to the commercial DynoMag® system that allows operation for f_{exc} ranging from 1 Hz up to 500 kHz. Moreover, the static bias field of 450 μ T is expected to modify $\chi(\omega)$, leading to a reduction in susceptibility amplitude and a decrease in τ_B [25].

IV. CONCLUSION

Overall, the presented results demonstrate the feasibility of MNP detection using the delta-E sensor as a proof-of-concept. However, in its current implementation, the system exhibits lower SNR performance and reduced bandwidth compared to the commercial DynoMag® system as well as previous SAW magnetoelastic sensor-based [18] and gradiometric coil-based approaches [7]. Further optimization of the transduction mechanism and operating point is expected to improve performance, thereby enabling future biomedical sensing applications based on magnetoelectric sensor technologies.

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